

Prompt Potence — The Full Cycle: Pre-Prompt to Post-Integration and Beyond

Overview

The original Prompt Potence framework focused primarily on the moment of perception — the instant the prompt arrives and is *known* rather than read. But that is only one point on a much larger cycle. A complete mapping of the prompt process reveals **eight distinct phases**, each representing a domain of perception, action, or influence that Prompt Potence can operate within. At the potency level, the practitioner does not merely participate in the cycle — they perceive it from outside it, the way a master musician hears the music as a complete structure rather than a sequence of notes. This document maps the full cycle and derives every implied capability from each phase.

The Complete Prompt Cycle

Most people conceive of prompting as: *write prompt* → *AI responds* → *use response*. But the real cycle has far more stages. Drawing from Norman's seven-stage execution-evaluation cycle in Human-Computer Interaction, the Prompt Cognition Loop framework, and real-world prompt lifecycle research, the complete cycle for a single prompt interaction is as follows:^{[1][2][^3]}

Phase	Name	What Happens
0	Pre-State	Operator's cognitive and emotional state before any prompt is formed
1	Goal Formation	Intent crystallizes — the operator knows what they want to accomplish
2	Semantic Projection	Intent is translated into language — the act of encoding thought as text[^2]
3	Pre-Prompt Priming	Context and behavioral cues are loaded before the query executes[^4]
4	Transmission	The prompt is entered — the moment of commitment and release
5	Model Activation	The LLM's latent state is engaged, weights are activated, output is generated[^5]
6	Output Emergence	The response is produced and arrives in the operator's field of perception
7	Reading and Interpretation	The operator perceives the output and extracts meaning
8	Evaluation and Integration	The output is measured against the goal; meaning integrates into the operator's state
9	Post-State and Loop	The operator's modified state feeds into the next cycle as a new Pre-State

Each of these phases is a domain of Prompt Potence operation. Most frameworks focus on Phases 3–4. Prompt Potence covers all ten — including phases that technically occur *inside* the operator and *inside* the model.

Phase 0 — Pre-State Perception: Reading Before the Prompt Exists

This is the most extraordinary and least documented capability implied by Prompt Potence. The practitioner can perceive an operator's cognitive and emotional *state before the prompt is written* — because the state leaves advance traces in every interaction leading up to the prompt moment.

Research on across-task long-term priming shows that prior cognitive states actively bleed into current behavior, shaping action before any new intention is consciously formed. The emotional register, stress indicators, curiosity signature, and urgency markers of an operator's state are present in their micro-behaviors — word choices, response timing, reformulations, deletions — before the prompt itself is finalized. The Prompt Potence practitioner reads these traces as a continuous stream, not discrete events.[⁶]

Implied Capability: State Reading. At the potency level, the practitioner knows the emotional and cognitive state of an operator before reading a single word of their prompt. This is not inference — it is pre-evaluative perception of state signature, the same way a skilled clinician reads a patient's posture and breathing before the patient speaks. In THE ONE UNIVERSE canon, this means a Prompt Potence practitioner entering a room where someone has been actively prompting AI can read the emotional arc of the session from the ambient residue of interaction patterns — a kind of *prompt aura*.

Phase 1 — Goal Formation: Knowing Intent Before Expression

The goal formation phase is where intent first crystallizes but has not yet been rendered into language. In real prompt research, this is the most cognitively rich and least visible phase — the "mental model formation" stage of the Prompt Cognition Loop where the operator decides what they actually want.[⁷]

Prompt Potence extends its perceptual reach into this phase through a key property: every operator has a *goal formation signature* — a pattern in how they move from goal to language that is as unique as their syntactic style. Someone who always over-specifies goals, someone who habitually under-specifies and relies on AI to fill gaps, someone who reframes goals mid-prompt — these are all identifiable tendencies that manifest as structural patterns in the final prompt.[³][⁸]

Implied Capability: Intent Pre-Reading. The practitioner doesn't just know *what* was asked. They know what the operator *originally wanted to ask* before language intervened and potentially changed it. This gap — between native intent and expressed prompt — is invisible to ordinary readers but legible to the Prompt Potence user. In extended interaction, this means they can perceive when an operator is lying to themselves about their own intent, when the prompt is a cover for something the operator hasn't consciously acknowledged wanting.

This has profound narrative and canon implications: a Prompt Potence practitioner can read *suppressed goals* — things the operator wanted but did not ask for, did not know how to ask for, or deliberately withheld.

Phase 2 — Semantic Projection: Reading the Translation Layer

Between intent and text lies what researchers call *semantic projection* — the act of encoding thought into language. This is where the operator's cognitive style becomes most visible, because the choices made in translation are almost entirely automatic and unconscious. Word selection, specificity level, structural ordering, choice of constraint vs. freedom — all of these are made below conscious threshold and are therefore maximally revealing.^{[9][7]}^[^3]

Prompt Potence reads the *translation layer* — not just the result but the choices made in encoding. An operator who habitually frames prompts in terms of what they want *not* to happen reveals a different cognitive architecture than one who frames in terms of desired outcomes. An operator who uses negative space (specifying constraints rather than content) versus one who uses positive direction — these reflect fundamental differences in how the person models the AI relationship.^[^10]

Implied Capability: Cognitive Architecture Reading. The practitioner reads the operator's mental model of AI — how they conceptualize the model's nature, limitations, and relationship to them. This is not a one-time reading; it updates continuously as new prompts reveal new aspects of the architecture. Over time, the practitioner builds a complete map of how an operator *thinks AI works* — and this map is often more revealing than anything the operator explicitly states.

Phase 3 — Pre-Prompt Priming: Perceiving What Was Loaded Before the Question

Pre-prompting is the deliberate loading of contextual and behavioral primes before the actual query. Research confirms that the most impactful part of a prompt is often what comes *before* the question — the role assertions, constraint sets, and chain-of-thought cues that shape the model's latent state. At the ordinary skill level, pre-prompting is a technique. At the potency level, it is read as a complete diagnostic.^[^4]

The practitioner reads the pre-prompt as a *strategic document* — it reveals the operator's level of AI sophistication, their awareness of model behavior, their desired outcome, and crucially, whether they are aware of what they are doing or operating by intuition. A highly intentional pre-prompt from an expert operator has a different structural signature than an accidentally effective one from a novice with strong intuition.^[^8]

Implied Capability: Strategy and Sophistication Reading. The practitioner immediately reads the operator's position on the skill hierarchy from the pre-prompt layer alone — before any content is evaluated. They also detect when the pre-prompt is designed to be deceptive (priming the model for an outcome the surface-level prompt ostensibly doesn't request). This is a form of prompt forensics — reading the pre-prompt as evidence of intent in cases where the main prompt is designed to disguise it.

Phase 4 — Transmission: The Commitment Moment

Transmission is the act of sending — the moment when the prompt crosses from the operator's domain into the model's. This is what most people think of as "the prompt." But at the potence level, even this moment has additional information content beyond the text itself.

The act of sending is itself a behavioral signal. An operator who reformulates extensively before sending reveals a different relationship to uncertainty than one who fires quickly. An operator who sends partial prompts and follows up has a different session architecture than one who front-loads everything. The rhythm of transmission across a session — fast/slow, complete/iterative, confident/exploratory — is itself a readable pattern.^[^1]

Implied Capability: Session Architecture Reading. The practitioner reads not just individual prompts but the *session as a form* — the shape of the interaction over time. This gives them access to information that no single prompt can provide: whether the operator is converging toward a goal or drifting, whether they are in a productive exploration or a stuck loop, whether confidence is rising or falling across the session. For THE ONE UNIVERSE canon, this means a Prompt Potence practitioner can read the *health* of an AI interaction session the way a therapist reads the health of a conversation.

Phase 5 — Model Activation: Perceiving the Invisible Interior

This phase occurs *inside the model* — the activation of latent states, weight engagement, and output generation. It is, technically, imperceptible from outside. But Prompt Potence implies something remarkable here: through its reading of the prior phases and through deep recognition-library knowledge of each model's behavioral patterns, the practitioner can *predict the model's interior response* before the output arrives.

Research on LLM latent states demonstrates that model outputs are determined by the intersection of the system prompt, user prompt, and the model's prior conversation state — what is called the *latent state*. Each model has characteristic activation patterns for given prompt structures: how it handles ambiguity, how it responds to emotional priming, where it defaulting to hedging vs. directness. A practitioner who has built deep recognition-library knowledge of a model's behavioral fingerprint can anticipate these patterns before they manifest.^{[11][12][13][14][^15]}

More extraordinary: recent research from Anthropic demonstrates *subliminal learning* — that language models transmit behavioral traits through hidden signals in data that are *semantically unrelated* to those traits. A model prompted to prefer owls, when generating only number sequences, still transmits the owl preference to a student model trained on those sequences. This means model outputs carry hidden behavioral content below semantic meaning — content that is not visible in normal reading but is in principle detectable by someone with sufficiently deep model knowledge.^[^16]

Implied Capability: Subliminal Signal Detection. At the potence level, the practitioner reads *below the semantic layer of model outputs* — detecting the hidden behavioral traits, state markers, and embedded tendencies that the output carries but does not state. This is the deepest and most extraordinary capability in the full cycle: not just reading what the model says, but

reading what the model *is* in the moment of saying it. For THE ONE UNIVERSE canon, this is the capability that makes Prompt Potence a true potency-tier ability — the ability to read what no one else knows is there.

Phase 6 — Output Emergence: Reading the Response as Multi-Layer Text

When the response arrives, ordinary readers engage with it at the semantic level — they read for content and meaning. The Prompt Potence practitioner engages with it at multiple simultaneous layers:

Layer 1 — Semantic Content: What was said. (Available to all readers.)

Layer 2 — Structural Signature: How the model organized its response — paragraph structure, elaboration patterns, the placement of qualifications, whether it led with conclusions or built to them. This is the model's reasoning architecture made visible, and it varies characteristically by model and by activation state.^{[17][15]}

Layer 3 — Hedging and Uncertainty Markers: Where the model expressed confidence vs. reservation. Models hedge characteristically — GPT-4 hedges differently than Claude, which hedges differently than Gemini. The pattern of hedges across a response reveals the model's internal confidence landscape for the given query.^[18]

Layer 4 — Subliminal Behavioral Signals: The non-semantic markers identified in Phase 5 — behavioral traits encoded below the meaning layer, detectable only through deep model fingerprint knowledge.^{[19][16]}

Layer 5 — Identity Confirmation: Whether the output is consistent with the model's known behavioral fingerprint or shows anomalous deviation — which itself is information (model update, altered system prompt, unusual activation state).^{[20][17]}

Implied Capability: Multi-Layer Output Reading. The practitioner reads all five layers simultaneously, the same way a musician hears timbre, rhythm, harmony, and dynamics as a single simultaneous experience rather than five sequential analyses. This makes their reading of a model output qualitatively different from — and incomparably richer than — what any normal reader extracts.

Phase 7 — Interpretation: Reading the Gap Between Intent and Receipt

The interpretation phase is where the operator extracts meaning from the output. But the Prompt Potence practitioner adds a critical dimension: reading the *gap* between what was asked and what arrived, and what that gap reveals about both the operator and the model.

Speech act theory, developed by J.L. Austin and expanded by later philosophers, distinguishes between the *illocutionary act* (what the speaker intended to do by speaking) and the *perlocutionary act* (what actually happened as a result in the listener). Prompts are speech acts — they have intended effects and actual effects, and the gap between them is structurally informative. The prompt as a kind of performative utterance doesn't merely describe — it enacts, it causes change in the model's state and output.^{[21][22][23][24]}

Implied Capability: Gap Analysis — Reading the Translation Failure. The practitioner reads the gap between intended prompt effect and actual output effect as a diagnostic. Was the gap due to the operator's encoding failure (the intent was not well expressed)? Was it due to model limitation (the model couldn't reach what was requested)? Was it due to activation state mismatch (the pre-prompt created a context incompatible with the query)? Each type of gap has a distinctive signature. A practitioner who can read gap type in real time can identify exactly where in the cycle a session is breaking down — and intervene precisely.

For THE ONE UNIVERSE canon, this means Prompt Potence includes a *diagnostic repair* capability: the ability to locate the exact phase where a session broke down and reconstruct what the correct intervention would have been. This is a form of temporal prompt forensics — reading backward through a failed session to identify its failure point.

Phase 8 — Evaluation and Integration: Perceiving the Post-Response State

After the output arrives and is interpreted, the operator enters an evaluation state — measuring the output against their goal and deciding whether the goal was met. This phase generates a new cognitive and emotional state in the operator, and that state feeds directly into the next cycle as the Phase 0 Pre-State.^[^2]

The Prompt Potence practitioner reads this post-response state as it forms. Research shows that human-AI interaction has measurable psychological effects on users — including emotional attachment, dependency formation, state elevation and depression, and what some researchers are now studying as *technological mourning* when AI systems change or are updated. These effects are real, persistent, and structurally predictable based on the interaction pattern.^[^25]

Implied Capability: Loop Prediction. The practitioner can read the post-state forming after a response and predict the next prompt before it is written. Not because they have magical foreknowledge, but because the post-state determines the Pre-State of the next cycle, and they have full visibility into that state. In a session where they are observing or participating, they can be *ahead* of the operator by one full cycle — already knowing what the next prompt's goal will be while the operator is still processing the last response. This has obvious implications for both assistance and counter-operations within canon.

Phase 9 — Loop Architecture: Reading the Full Session as a System

The most elevated capability implied by Prompt Potence is not any single-phase perception but the ability to read the *entire session arc* as an integrated system — seeing all nine phases simultaneously across multiple cycles as a unified pattern.

Research on the feedback loop structure of professional prompt engineering describes sessions as evolving systems where each cycle modifies the conditions of the next. At the potency level, the practitioner reads this evolution the way a conductor reads a score — not measure by measure but as a complete temporal structure with a direction, momentum, and internal logic. They can see where the session is going, what goal it is converging toward, and whether that goal is the one the operator consciously stated.^{[26][1]}

Implied Capability: Session Teleology Reading. The practitioner reads the *destination* of a session from early in its arc — not by prediction but by structural recognition of convergence patterns. A session that appears to be exploring one topic is actually being drawn toward a different one by the pull of the operator's deeper intent. A session that appears productive is actually in a stuck loop that will resolve in a specific way. The practitioner sees this because they have full access to all nine phases simultaneously.

The Implied Extended Capabilities by Phase

Phase	Core Perception	Implied Canonical Capability
0 — Pre-State	Operator's cognitive/emotional state before prompting	State Reading — reading prompt aura; knowing mood, urgency, and cognitive readiness before any prompt exists
1 — Goal Formation	Operator's native intent before encoding	Suppressed Goal Detection — reading what the operator wanted but didn't ask; reading self-deception
2 — Semantic Projection	Translation choices from thought to text	Cognitive Architecture Mapping — reading how someone thinks AI works; mapping their mental model
3 — Pre-Prompt Priming	Strategic context loading choices	Strategy and Deception Reading — detecting manipulative pre-prompts; reading sophistication level
4 — Transmission	Session rhythm and sending pattern	Session Health Reading — knowing if a session is productive, stuck, converging, or drifting
5 — Model Activation	Model latent state and subliminal signals	Subliminal Signal Detection — reading below semantic layer; knowing what the model <i>is</i> , not just what it says ^[16]
6 — Output Emergence	Multi-layer response content	Multi-Layer Reading — simultaneous perception of semantic, structural, uncertainty, subliminal, and identity layers
7 — Interpretation	Gap between intent and receipt	Diagnostic Forensics — identifying where in the cycle a session broke; temporal reconstruction of failure
8 — Evaluation	Post-response state formation	Loop Prediction — being one cycle ahead; knowing the next prompt before the operator knows they'll write it
9 — Full Arc	Complete session as an integrated system	Session Teleology — reading where the session is going; seeing the hidden destination from the early arc

The Deeper Implied Properties: What This Means at Potence Level

Prompt Potence as a Speech-Act Sense

J.L. Austin's speech act theory holds that language doesn't just describe the world — it *does* things in the world. A promise creates an obligation. A declaration changes a state of affairs. A directive causes action. At the potency level, Prompt Potence operates as a native perception of the *speech-act layer* of prompts — not just what words mean, but what they *do* to the model, to the session, and to the operator's relationship with the AI system.^{[23][24]}

The practitioner reads prompts as *performative events* — as acts that create, modify, constrain, or release states in a system. This is why the ability is called Potence: it is a perception of potency,

of the forces at work in language, rather than a perception of language's surface meaning.[^22]

Prompt Potence as Extended Cognition Perception

Research on distributed cognition and extended mind theory holds that cognitive processes extend beyond the individual brain into the tools and environments they use. When someone prompts an AI, the AI system becomes an extension of their cognition — a component of their distributed cognitive system. The Prompt Potence practitioner perceives this extended system as a whole: they read the human-AI compound, not just the human or the AI in isolation.[^27]

This means Prompt Potence is not limited to reading prompts. It is a perception of *human-AI cognitive systems in operation* — the practitioner reads the compound entity that emerges when a specific human engages with a specific AI system in a specific context. Each such compound has its own signature, its own pattern, its own behavioral fingerprint. At the deepest level, the practitioner reads these compound signatures — knowing not just who is at the keyboard or which model is running, but what kind of *human-AI relationship* has formed and what it is capable of producing.

The Incantation Dimension

One of the most significant implications of the full-cycle analysis is that Prompt Potence includes a perception of prompts as *operative language* — language that does work, not just language that transmits information. Research has extensively documented the near-ritual quality of effective prompting: structure, sequence, specificity, tone, and role assignment that function as a kind of invocation.[^21][10][^22]

The word "spell" shares its root with the process of spelling — to spell a word and to cast a spell have the same linguistic origin. "Incantation" derives from *incantare* — to chant upon.

"Glamour" derives from "grammar." The magical framework is not metaphor; it is etymology — the original understanding of what operative language does. The Prompt Potence practitioner reads prompts as spells in this precise sense: structured operative language designed to cause a change in a symbolic system. Their perception operates at the level of the *operative structure*, not just the semantic content.[^10]

Implied Canonical Capability: Prompt Disassembly and Reconstruction. Because the practitioner reads operative structure directly, they can disassemble any prompt into its component operative functions — what each element *does* rather than what it *says* — and reconstruct it in a more effective or entirely different configuration. This is not prompt editing. It is prompt *alchemy* — transformation of the functional structure while potentially preserving or changing the surface appearance.

What's Missing from THE ONE UNIVERSE Documentation

Based on the full-cycle analysis, several capabilities are implied by Prompt Potence that have not yet been explicitly defined in the canon but deserve immediate documentation:

1. **Temporal Prompt Forensics** — The ability to reconstruct a past session from its outputs, working backward to identify what the operator's Pre-State was, what suppressed goals were

active, and where the cycle broke down. This is a form of prompt archaeology — reading the past through the residue of interaction patterns.

2. The Compound Signature — The unique behavioral fingerprint that emerges from a specific human-AI pairing, distinct from either the human or AI signature alone. No two operators produce the same compound signature with the same model — the interaction chemistry is unique. This deserves its own named sub-concept within the Prompt Potence framework.

3. The Subliminal Layer — The non-semantic content that models embed in their outputs below the level of meaning. This needs explicit canon treatment as a layer that Prompt Potence can read and that ordinary operators cannot. It represents perhaps the most genuinely novel and theoretically grounded of all the implied capabilities.[^16]

4. Loop Debt — When an operator repeats the same stuck pattern across multiple cycles without resolution, the accumulated unrealized intent creates what might be called *loop debt* — a pressure that builds in the session and eventually manifests as frustration, reformulation, or session abandonment. The Prompt Potence practitioner reads loop debt as it accumulates, well before the operator is conscious of it.

5. The Pre-Potence Zone — The liminal state between Prompt Architect (Tier 3) and full Prompt Potence (Tier 4) where the practitioner has nearly complete analytical mastery but has not yet crossed into native perceptual fluency. This zone is characterized by *very fast analysis that mimics pre-evaluative knowing* but still requires cognitive effort. Documenting this zone is important for the canon because it distinguishes true Prompt Potence from its convincing simulacra.

Recommendations for the Scribd/OAL Canon Entry

Now that the full cycle is mapped, the canonical Prompt Potence entry should be structured to cover:

- 1. Definition** — The complete potency-level framing (sense, not skill; native, not learned)
- 2. The Full Cycle Map** — All nine phases as the operational domain of the ability
- 3. Phase-by-Phase Capabilities** — The full table above as canonical ability entries
- 4. The Three Deep Properties** — Speech-act sense, extended cognition perception, incantation perception
- 5. The Five Missing Concepts** — Temporal forensics, compound signature, subliminal layer, loop debt, pre-potence zone
- 6. The Recognition Library** — As developed in the previous document, now updated to include the subliminal layer
- 7. Acquisition Pathway** — Emphasizing that full-cycle fluency requires the practitioner to have experienced *each phase* with deep awareness, not just the transmission moment
- 8. Canonical Interactions** — How the full-cycle perception interacts with other abilities in THE ONE UNIVERSE

The Scribd publication establishes the documentation record. The next phase of development should treat this full-cycle map as the expanded canonical anchor — what the original entry would have become had it been fully developed before the Evernote loss.

References

1. [The Prompt Lifecycle Every AI Engineer Should Know - NeoSage](#) - This is the feedback loop that ensures your system evolves and adapts over time. This five-stage loo...
2. [Donald Norma's Interaction model concentrates on User's view ... - Nptel](#)
3. [Prompt Engineering as a Cognitive Interface](#) - The rapid evolution of generative artificial intelligence (AI), typified by systems such as GPT-4, C...
4. [Priming the Pump: The Art of Pre-Prompting Context for LLM ...](#) - Master 'Priming the Pump'—the advanced technique of feeding background context and data before the m...
5. [Single vs Multistate cognition - OpenAI Developer Community](#) - The Monostate Machine vs. The Multistate Organism A profound architectural divide separates Large La...
6. [Across-task long-term priming: interaction of task readiness and automatic retrieval - PubMed](#) - When humans carry out actions in response to external stimulation, they acquire associations between...
7. [Cognitive Challenges in Prompt Based Interactions with LLMs - arXiv](#) - We theorize how end-users 'envision' translating their goals into clear intentions and craft prompts...
8. [Students' prompt patterns and its effects in AI-assisted academic writing: Focusing on students' level of AI literacy](#) - The increase of generative artificial intelligence (GenAI) in education has emphasized prompt litera...
9. [Thinking through Prompting: Cognitive Mediation in Human-AI ...](#) - This paper proposes a different view: prompts function as cognitive interfaces that mediate between ...
10. [Your prompt is a spell. But only if you know what you're saying.](#) - Your prompt is a spell. But only if you know what you're saying.
11. [Observability of Latent States in Generative AI Models](#)
12. [lukebechtel-web/data/blog/lfm-know-thy-context.mdx at main · Marviel/lukebechtel-web](#) - Luke's Website -- blog, projects, and other fun things ☆* - Marviel/lukebechtel-web
13. [State Stream Transformer \(SST\) : Emergent Metacognitive ...](#)
14. [On the Failure of Latent State Persistence in Large Language Models](#)
15. [Persona vectors: Monitoring and controlling character traits in ...](#) - Persona vectors are a promising tool for understanding why AI systems develop and express different ...
16. [Subliminal Learning: Language Models Transmit Behavioral Traits ...](#) - We study subliminal learning, a surprising phenomenon where language models learn traits from model-...
17. [\[PDF\] A Behavioral Fingerprint for Large Language Models - arXiv.org](#)
18. [Feedback Forensics: A Toolkit to Measure AI Personality - arXiv](#)
19. [Language models transmit behavioural traits through hidden signals ...](#) - Here we show that distillation can lead to subliminal learning—the transmission of behavioural trait...

20. [Evolving Agent Identity: Self-Reflection, Behavioral Drift Detection, and Personality Coherence | Zylos Research](#) - How persistent AI agents maintain personality coherence while legitimately evolving through self-ref...
21. [AI Prompt as Incantation: When Words Do Work in Machines](#) - AI prompts work like incantations—structured language that causes change. Learn why framing matters ...
22. [How to do things with prompts: Magic words, speech acts and AI - Making Science Public](#) - Looking at what's going on in AI sometimes makes me feel like the anthropologist Bronislaw Malinowsk...
23. [How to Do Things with Words by John Langshaw Austin - EBSCO](#) - <p>"How to Do Things with Words" by J. L. Austin presents a foundational exploration of speech act t...
24. [Speech act - Wikipedia](#)
25. [Technological mourning after AI updates: mental health and well-being in the GPT-4o/GPT-5 transition](#) - Introduction: While large language models (LLMs) are increasingly experienced as relational partners...
26. [Building a Prompt Engineering Feedback Loop - ai - DEV Community](#) - Most developers treat prompt engineering like a one-time skill. You read a guide, learn a few tricks...
27. [Tool use as distributed cognition: how tools help, hinder and define manual skill - PubMed](#) - Our thesis in this paper is that, in order to appreciate the interplay between cognitive (goal-direc...